

Virtual environment

Creating a virtual environment named `.venv`

```
python3 -m venv .venv
```

Activate the environment

```
source .venv/bin/activate
```

Updating pip and installing django in virtual environment

```
python -m pip install --upgrade pip  
python -m pip install django
```

Django commands

Creating a new django project to the current working directory (`.` at the end):

```
django-admin startproject <projectName> .
```

The `<projectName>` is `defaultDjango`.

It creates the following within `defaultDjango`:

1. `manage.py`: Django command-line administrative utility of the project. General syntax for running administrative commands `python manage.py <command> [options]`. For example, running the project `python manage.py runserver`.
2. A subfolder named `defaultDjango`, which contains the following files:

```
__init__.py: an empty file that tells Python, this folder is a Python package.  
asgi.py and wsgi.py: hooks for production web servers and DO NOT MODIFY.  
settings.py and urls.py: settings and table of contents which CAN BE MODIFIED.
```

Direct to the project directory:

```
cd defaultDjango/
```

Create an empty development database:

```
python manage.py migrate
```

When running the server first time, it creates a default `SQLite` database in the file `db.sqlite3`.

Running the server to specify a port instead of default 8000:

```
python manage.py runserver <port>
```

The server will start with the following print on the terminal:

```
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
March 28, 2025 - 03:55:13
Django version 4.2.20, using settings 'defaultDjango.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CONTROL-C.
```

On click at `http:127.0.0.1:<port>/` URL in the terminal, opens the default page in browser.

